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Two-Way Joint Source-Channel Coding with a Fidelity Criterion

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Abstract

Consider a source, $\{X_i, Y_i\}_{i=1}^{\infty}$, producing independent copies of a pair of jointly distributed RVs. The $\{X_i\}$ part of the process is observed at some location, say A, and is supposed to be reproduced at a different location, say B, where the $\{Y_i\}$ part of the process is observed. Similarly, $\{Y_i\}$ should be reproduced at location A. The communication between the two locations is carried out across two memoryless channels in K iterative bi-directional rounds. In each round, the source components are reconstructed at the other locations based on the information exchanged in all previous rounds and the source component known at that location, and it is desired to find the amount of information that should be exchanged between the two locations in each round, so that the distortions incurred (in each round) will not exceed given thresholds. We first derive a single-letter characterization of achievable rates for a pure source-coding problem with successive refinement. Then, for a joint source-channel coding setting, we prove a separation theorem, asserting that in the limit of long blocks, no optimality is lost by first applying lossy (two-way) successive-refinement source coding, regardless of the channels, and then applying good channel codes to each one of the resulting bitstreams, regardless of the source.

Index terms - two-way communication, source coding, joint source-channel coding, successive refinement, channel capacity.

1 Introduction

In recent years, communication over distributed networks has been developed greatly. Intensive research was carried out in both theoretical and practical directions, analyzing

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